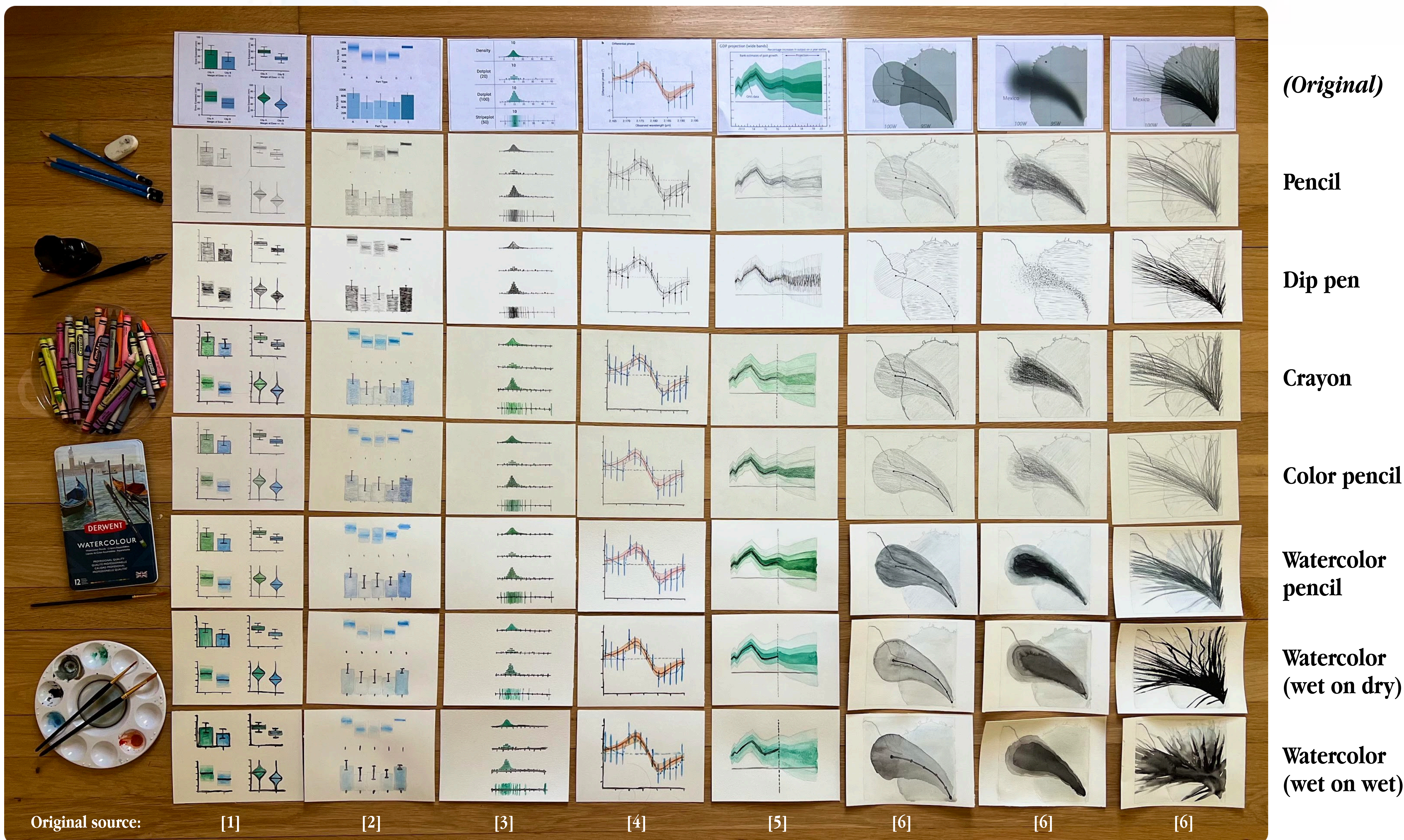


# ARTISTIC INSPIRATION FOR UNCERTAINTY VISUALIZATION TECHNIQUES

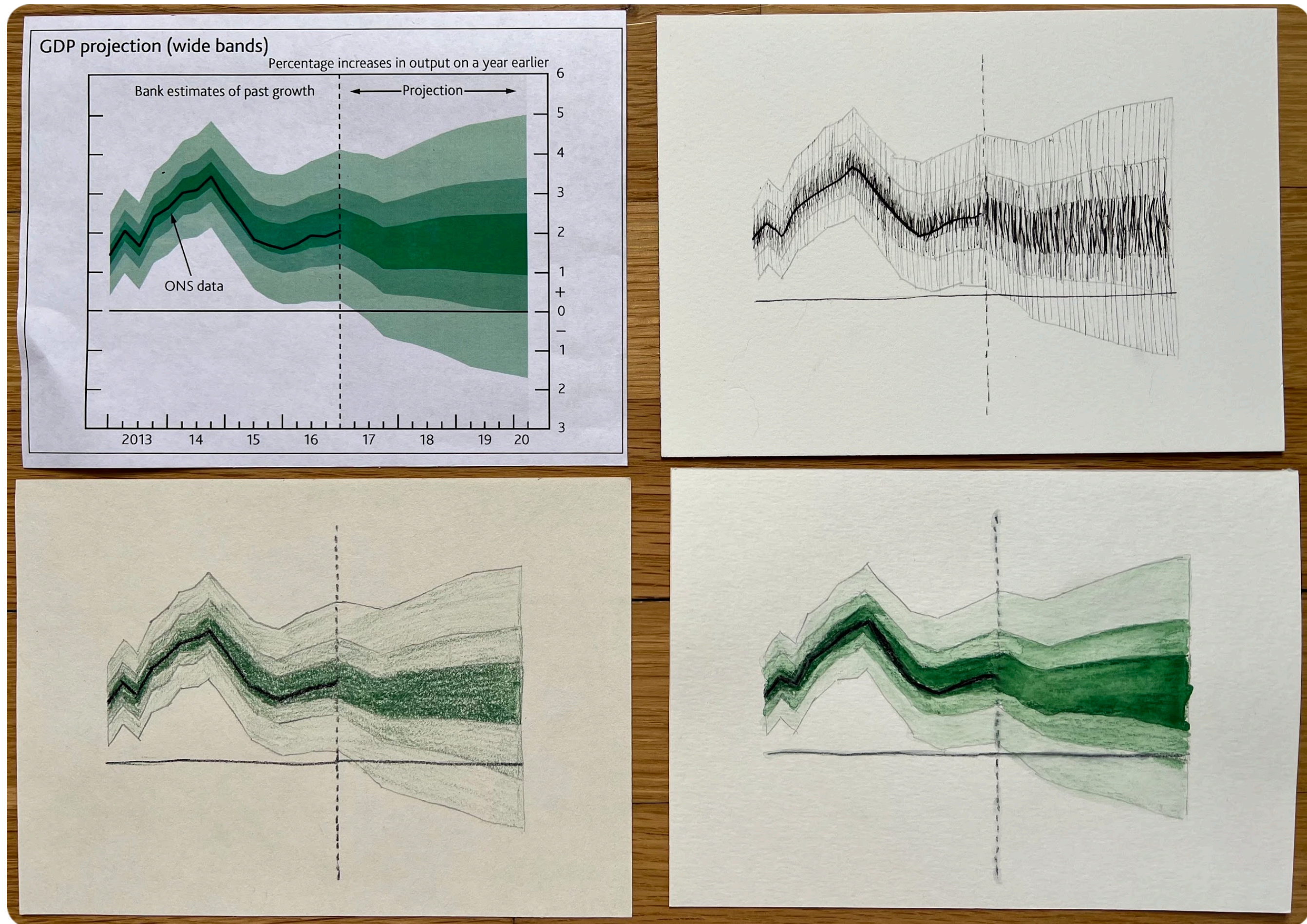
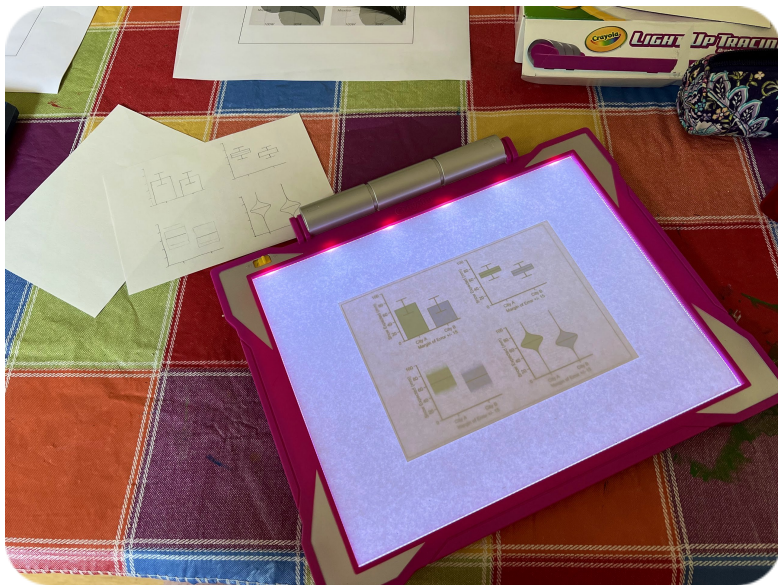


Michelle A. Borkin

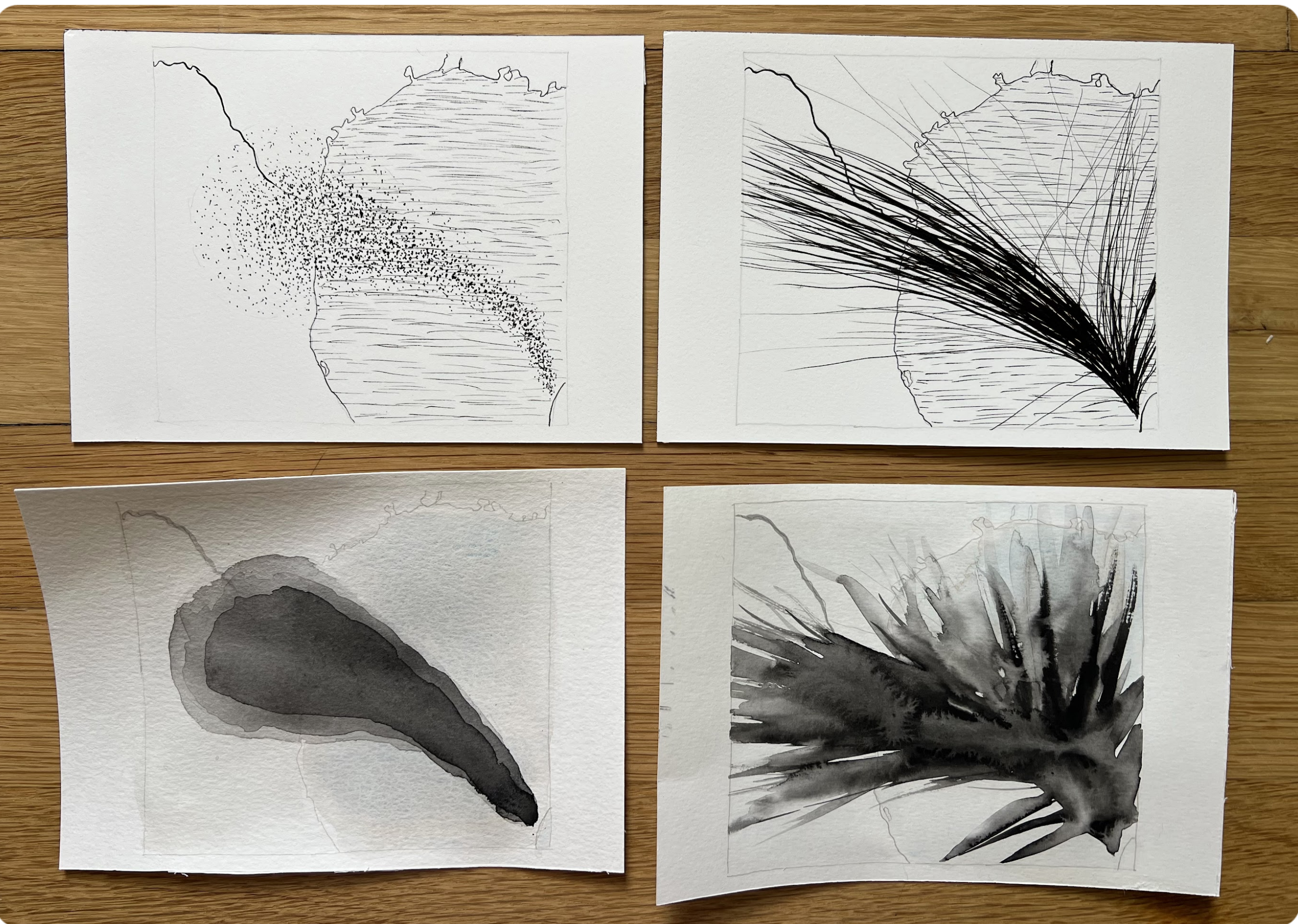


**ABSTRACT:** Uncertainty visualizations from error bars to distributions to fuzzy spatial boundaries on maps are ever-present and important for accurate data and model interpretation. But could these visualizations be interpreted differently when presented in different visual styles? Are there undiscovered ways to represent the visual channels of fuzziness and transparency, and their combination? To address these questions this work explores artistic drawing and painting techniques to more deeply understand the role and impact of aesthetic styles and visual channels on uncertainty visualizations. This work contributes 56 hand drawn and painted reproductions of existing uncertainty visualizations with an accompanying reflection and analysis. The collection visually demonstrates the role of aesthetics, visual perception principles, and artistic techniques for fuzziness and transparency in uncertainty visualization as well as avenues for new encoding techniques.

**PROCESS:** The reproductions were created by placing a print-out of the original figure on a light table to trace onto a 5" x 7" cut piece of paper. It took on average for each reproduction ~10 minutes to trace and ~15-45 minutes to draw or paint. The creation of the 56 reproductions was completed over several weeks.



**Original print of a fan chart with reproductions in dip pen, colored pencil, and watercolor pencil.** The vertical hatch marks are one example perceptual principles at play demonstrating an illusion of jitter or movement due to the high spatial frequency of the pattern.



**Comparison of the dip pen and watercolor (wet on wet technique) for the blurred hurricane projected track visualization and corresponding ensemble plot.** Different representations imitate the original context (e.g., hatching pattern looks like waves in ink, and the watercolor blur is akin to clouds), and evoke different emotions and interpretations based on the aesthetics. The watercolor reproductions in particular due to the randomness and interplay of the fuzziness and transparency combine the visual encoding channels for uncertainty.

## UNCERTAINTY VISUALIZATION SOURCES:

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**What are your thoughts? Let me know!**  
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